A MONOMORPHISM OF ALGEBRAIC SPACES OVER A FIELD IS EITHER EMPTY OR THE SPECTRUM OF THE FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof combines the dense-scheme result for separated spaces with the scheme case; both cases of the scheme statement are then converted back to Z.

Editorial note. The four references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- A monomorphism is separated and separatedness over a separated base gives separatedness. Lemmas lemma-monomorphism-separated and lemma-separated-over-separated, used at proof lines 2006-2009. Makes Z a separated algebraic space.
- A separated algebraic space has an open dense subspace which is a scheme. Properties of Spaces, Proposition spaces-properties-proposition-locally-quasi-separated-open-dense-scheme, used at proof lines 2009-2012. Produces the open dense Z' to which the scheme statement applies.
- The scheme case for a monomorphism towards a field. Schemes, Lemma schemes-lemma-mono-towards-spec-field, used at proof lines 2013-2015. Gives Z' empty or isomorphic to Spec(k).
- Density and monomorphismhood force Z' = Z in the isomorphism case. used at proof lines 2016-2019. Excludes intermediate cases.

Source reference. Original lemma 042N: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5334–5334. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 04ZH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6465–6465. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03DP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4428–4428. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original proposition 06NH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8536–8536. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let k be a field and let $Z \to \mathrm{Spec}(k)$ be a monomorphism of algebraic spaces over S. Then either $Z = \emptyset$ or $Z = \mathrm{Spec}(k)$.*

Proof. By Lemmas 042N and 04ZH we see that Z is a separated algebraic space. Hence there exists an open dense subspace $Z' \subset Z$ which is a scheme, see Properties of Spaces, Proposition 06NH. By Schemes, Lemma 03DP we see that either $Z' = \emptyset$ or $Z' \cong \mathrm{Spec}(k)$. In the first case we conclude that $Z = \emptyset$ and in the second case we conclude that $Z' = Z = \mathrm{Spec}(k)$ as $Z \to \mathrm{Spec}(k)$ is a monomorphism which is an isomorphism over Z'. □